

Calibrating the Decision, Not the Model: Per-Class Thresholds for Training-Free Open-Vocabulary Remote-Sensing Segmentation

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Abstract

Training-free open-vocabulary segmentation pipelines built on SAM 3 gate their predictions with a single confidence threshold and assign everything below it to a **background** class. On LoveDA this discards 29.68% of all real-class pixels (323,184,908), with silence outnumbering misclassification 3:1. We show the single threshold is the wrong *shape* for the problem, and that two corrections recover a substantial part of what it costs — one threshold per class, and one multiplier per class applied before the argmax — both fitted on 200 labelled tiles with no weights trained, the same supervision the baseline already spends tuning its one threshold. Per-class thresholds give $+1.18 \pm 0.45$ mIoU on LoveDA, $+1.34 \pm 0.39$ on UAVid, and $+2.51 \pm 0.34$ on a CLIP-based competitor at its own operating point, so the correction composes with prior work rather than competing with it. The multiplier reaches errors no threshold can, because it changes which class wins: Potsdam’s **tree** gains $+21.17$ IoU where thresholds move it $+0.32$. All results are verified in the unmodified evaluation pipeline. We then report a finding that limits our own: **the hand-written prompt list every training-free method inherits is a larger lever than the method**. Replacing one word with a dataset’s own class name is worth $+3.53$ mIoU on UAVid — more than our complete method earns on LoveDA — and removes 64% of the multiplier’s gain there, so we restate that dataset’s contribution as $+3.22$. The same rule applied to Potsdam *costs* -2.71 mIoU. No lookup rule exists, and no compared method reports its prompts. Finally, eleven label-free rules fail to reach the oracle bound, four of which provably estimate per-class precision: the optimal threshold solves a *coupled* multi-class objective that no per-class statistic can express, so the labelled calibration set is irreducible.

1 Introduction

A training-free open-vocabulary segmenter is handed a list of class names and asked to label every pixel without seeing an annotation. The dominant recipe for remote

sensing scores each class independently, takes an argmax, and then applies one global confidence threshold: pixels below it become **background**. The threshold is not incidental. It is tuned per dataset by its authors, on labelled data, and published as part of the configuration — 0.5 on LoveDA, 0.1 on OpenEarthMap, 0.3 on UAVid.

We begin from what that one number costs. Reproducing the strongest published SAM 3 pipeline at 47.38 mIoU on LoveDA against its reported 47.4, we measure that 29.68% of every pixel carrying a real land-cover label is assigned to **background** — 323,184,908 pixels — and that this silence outnumbers outright misclassification 3:1. The obvious remedy fails: sweeping the threshold down to 0.1 recovers two thirds of that mass at a cost of 1.73 wrong pixels for every right one, and **loses** mIoU.

The claim. A single confidence threshold is the wrong shape for a multi-class decision, because different classes are wrong in different directions. **water** on LoveDA is right 89.5% of the times it fires and finds only 54.7% of what is there; **road** is the reverse. One number cannot serve both, and the fitted values span $0.175 - -0.675$ to 0.675 where the published one is 0.5 for all of them. Fitting one threshold per class on 200 labelled tiles — **no weights trained, and the same protocol the baseline already uses for its single threshold** — recovers $+1.18 \pm 0.45$ mIoU.

Per-class thresholds are provably the *complete* family of corrections available after the argmax (Sec. 5). That result is what motivates the second lever: if nothing more can be done after the argmax, the remaining freedom is *at* it. We fit one multiplier per class applied to the scores before the argmax, with the threshold still reading the raw score, and it reaches an error mass no threshold can touch — a pixel whose correct class lost the argmax cannot be recovered by lowering that class’s bar.

And a finding that limits our own. Every method in this literature inherits a hand-written list of class prompts, and none report how it was chosen. We tested two, each with the prediction registered before the run. Replacing UAVid’s **vegetation** — a word that also

names trees — with the dataset’s own term *low vegetation* is worth +3.53 mIoU and removes 64% of our second lever’s gain there. The identical rule applied to Potsdam costs -2.71. **The vocabulary is the largest single lever we measure, its direction is not predictable, and it is unreported.**

Contributions.

1. **A method, and a completeness result that bounds it.** Per-class thresholds give $+1.18 \pm 0.45$ on LoveDA, +1.34 on UAVid and +2.51 on a CLIP pipeline, all verified end to end. Any per-class monotone recalibration composed with a global threshold reduces to this rule exactly (Sec. 5).
2. **A second lever, motivated by that bound.** A per-class scale before the argmax adds $+4.86 \pm 0.35$ on Potsdam, +1.16 on LoveDA and +5.89 on UAVid over thresholds alone. It reaches Potsdam’s **tree**, which has the largest precision–recall gap we measure (+54.7) and which thresholds move +0.32.
3. **A larger lever than either, and a correction to our own result.** One word is worth +3.53 on UAVid and +4.94 on LoveDA against +2.32 for our complete method; the same rule costs -2.71 on Potsdam (Sec. 7).
4. **A reporting recommendation.** We observe a catch-all class moving the headline mIoU in both directions and recommend reporting mIoU over the real classes beside it.
5. **A bound with its reason.** eleven label-free rules fail; four provably estimate per-class precision, which we had supposed was the missing quantity. It is not.

2 Related work

Dense open-vocabulary segmentation from CLIP begins with MaskCLIP [10] and continues through SCLIP [9], ClearCLIP [5] and ProxyCLIP [6], each removing an obstacle between CLIP’s image-level training and pixel-level use. SegEarth-OV [7] adapts that line to remote sensing with a feature upsampler, and SegEarth-OV3 [8] replaces the stack with SAM 3 [1], fusing a semantic and an instance head and gating by a presence score. That pipeline is our baseline and we reproduce it.

ConInfer [2] is the nearest competitor: it fits a per-scene visual prior over DINOv3 features and fuses it with a VLM posterior, and reports gains over SegEarth-OV on the same benchmarks. It is orthogonal to us — it changes the *scores*, we change the *decision rule* applied to them — and we show the two compose.

Our corrections are recalibration in the sense of Guo *et al.* [3] and Kull *et al.* [4], but applied to a decision rather than to a probability: we never ask the scores to be calibrated, only that the rule reading them be class-aware.

What we add to that literature is a completeness argument for the post-argmax family and a measured demonstration that the remaining freedom lies at the argmax itself.

3 Setup

We evaluate on four segmentation benchmarks and one competing pipeline. LoveDA (1669 validation tiles) and OpenEarthMap (384 of the official 500) are land-cover benchmarks; Potsdam is ISPRS aerial imagery at **5 cm** ground sampling distance; UAVid is low-altitude drone video, a different sensor and viewpoint from the other three. We reproduce the published baseline on each: 47.38 against 47.4 on LoveDA, 57.83 against 57.8 on Potsdam, 44.19 against 42.9 on OpenEarthMap, and 56.86 against 54.7 on UAVid. The last differs by +2.16 under identical code, configuration, vocabulary and thresholds, and we report both figures rather than explain it away (Sec. 9).

Two properties of UAVid change the protocol. Its validation split is 7 drone flights of 10 *consecutive* frames, so a frame-level cross-validation places near-duplicates of every evaluation frame in the calibration set: it reports +1.05 where the flight-grouped answer is +0.51, **a leak worth +0.54 mIoU that flips two of five folds**. Its distributed training split further ships 400 flipped and shifted copies among 600 frames. All UAVid results here group folds by flight and exclude the augmented copies. We report both because neither is visible in the directory structure and both produce a clean, plausible, wrong number.

Protocol. Thresholds and scales are fitted by coordinate ascent on a calibration subset and evaluated on disjoint tiles, with the published-threshold baseline recomputed on the same held-out pixels. The fit maximises mIoU over the real classes only; full mIoU is still what we report. Every headline is verified by running the fitted parameters through the unmodified evaluation pipeline, not only through the cached statistic used to search.

4 What the baseline discards, and what governs it

How much real land cover a presence-gated pipeline discards is not a property of the model. It is ordered by how much of the scene the dataset’s catch-all class covers (Table 1). A catch-all gives the model a plausible answer everywhere, so a strong runner-up carries no information: on LoveDA nine tested signals reach a best AUC of 0.622 against a 0.53 floor, while on OpenEarthMap the runner-up score separates the residual at 0.913.

Label design, not the model, governs the background residual

A catch-all ‘background’ class gives SAM 3 a plausible answer everywhere: the residual is large and indistinguishable from genuine background. Where the vocabulary covers the scene, it is small and easily detected. Panel ① is the cause; ②–⑥ are consequences, each moving as it predicts. $n = 2$ datasets — the direction is consistent, the functional form is not claimed.

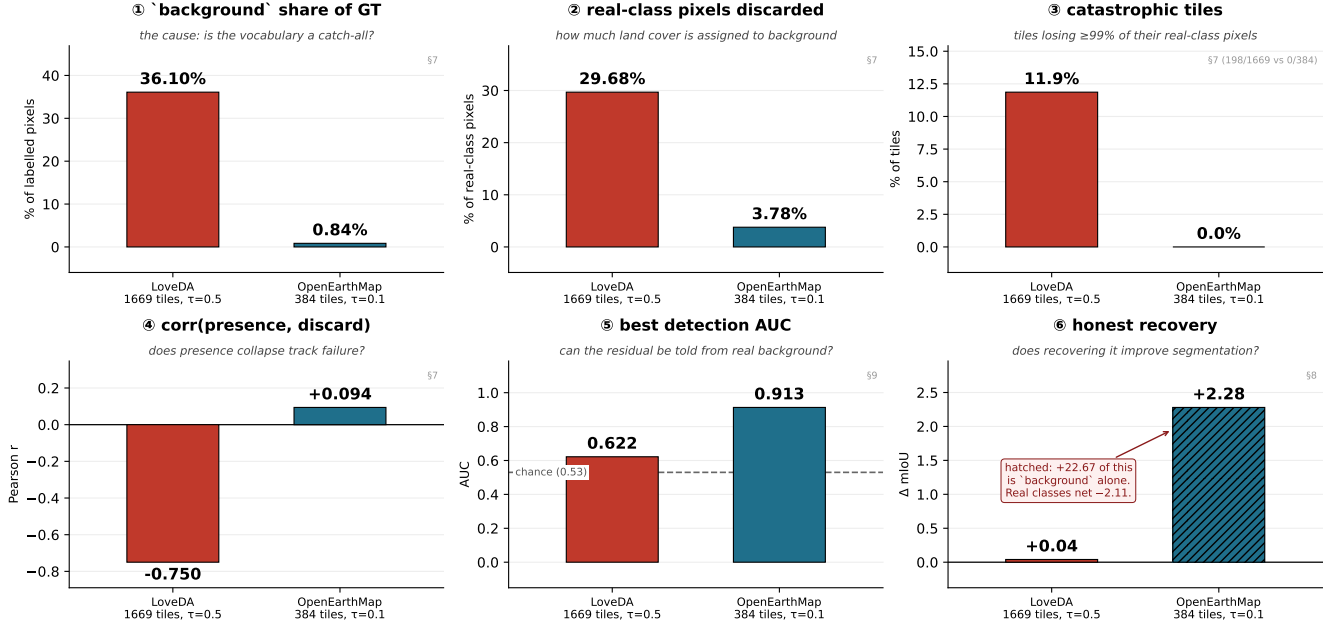


Figure 1: The residual’s size, its detectability, and the presence-score correlation all move with the catch-all class’s share of ground truth. The mechanism is established by intervention, not correlation: because that share is set by the vocabulary, we manipulate it directly with class count controlled.

Table 1: The residual is ordered by the catch-all’s share of ground truth, across four datasets. *catch-all* is that class’s share of ground truth; *discarded* is the share of real-class pixels assigned to it. The Potsdam row was predicted from ground-truth masks alone and registered before the pipeline was run.

	catch-all	discarded
OpenEarthMap	0.84%	3.78%
Potsdam	4.29%	4.68%
UAVid	16.16%	6.81%
LoveDA	36.1%	29.68%

Because the catch-all’s share is set by the *vocabulary*, we can intervene on it rather than stratify. Merging real classes into the catch-all at increasing doses, against a control that merges the same classes into *each other* so class count moves identically, drives detectability $0.913 \rightarrow 0.590$ while the control moves only to 0.855 (the supplementary material). A second arm localises the effect to the label space rather than the prompt list.

Recovery is not the answer. Having measured the residual, we tried to recover it and report the failure: a region-level labeller reaches +0.04 mIoU on LoveDA, and its apparent +2.28 on OpenEarthMap is entirely the catch-all ceasing to be over-predicted (+22.67 IoU) while

the real classes lose -2.11. **The residual is real and recovering it after the fact does not work.** What does work is not making the decision wrongly in the first place.

5 Two levers

Lever one: a threshold per class. The baseline predicts $\hat{c} = \arg \max_c s_c$ and emits **background** when $s_{\hat{c}} < \tau$. We replace the scalar with a vector: emit **background** when $s_{\hat{c}} < \tau_{\hat{c}}$. Nothing else changes, and with a constant vector the rule is bit-identical to the baseline.

This is the complete family of corrections available after the argmax. Let g_c be any per-class strictly monotone map applied to the scores, followed by a global threshold t . Because g_c is monotone, $g_c(s) \geq t$ holds exactly when $s \geq g_c^{-1}(t)$, so the composition is the per-class rule with $\tau_c = g_c^{-1}(t)$. Any post-argmax recalibration — temperature, vector scaling, isotonic — therefore reduces to choosing τ_c , and cannot beat choosing it directly.

The objective separates, so the fit is exact. For a fixed argmax, a pixel predicted c either clears τ_c or becomes the catch-all; it can never become a different real class. So TP_c , FP_c and FN_c depend on τ_c alone and the real-class objective is a sum of single-variable terms. Coordinate ascent is therefore **exact rather than**

Table 2: Both levers, across four datasets and a second pipeline. Δ is over the published threshold on identical held-out pixels. Catch-all-excluded mIoU is reported beside the headline throughout, for the reason in Sec. 6.

	lever 1	lever 2	excl. catch-all
LoveDA	+ 1.18	+ 1.16	+1.38 / +1.36
Potsdam	+0.59	+ 4.86	+1.04
UAVid	+ 1.34	+ 5.89	—
OpenEarthMap	+0.16	—	+ 1.75
ConInfer (CLIP)	+ 2.51	-0.10	+1.94

greedy — one sweep per class is the global optimum — and costs $N \times 201$ evaluations rather than 201^N . We verify this at exactly 0.000000 over 19 class sweeps on three datasets.

Lever two: a multiplier before the argmax. The completeness result holds only *given* the argmax, which points at the remaining freedom. We predict $\hat{c} = \arg \max_c w_c s_c$ and keep the pixel when $s_{\hat{c}} \geq \tau_{\hat{c}}$ — **the argmax reads the scaled score, the threshold reads the raw one**. That asymmetry is what confines w to the decision: a scale applied after the argmax would be monotone and merely reparameterise τ . Only ratios of w matter, so we renormalise to geometric mean one after each pass. (w, τ) are fitted jointly on the same tiles, and nothing is trained.

This reaches a mass no threshold can. A pixel whose correct class *lost* the argmax is not below that class’s bar — it is behind another class — and lowering its threshold returns it wearing the wrong label.

6 Results

Thresholds. Five-fold cross-validation gives $+1.18 \pm 0.45$ mIoU on LoveDA with every fold positive and a worst fold of +0.80, capturing 81% of an oracle bound of +1.46. The gain is entirely land cover: the catch-all moves -0.01 IoU. It is driven by **water**, which gains +6.78 IoU alone at a fitted threshold of 0.195 against a global 0.5. **The headroom is smaller than it looks, which makes the gain larger than it looks:** a fully supervised SegFormer-b0 trained on the whole LoveDA training set reaches 50.0, so **44% of the gap between training-free and fully supervised is closed by 200 labelled tiles and no trained weights**.

It is not backbone-specific, but it is dataset-specific. Applied unchanged to ConInfer at *its* published operating point, the same rule adds $+2.51 \pm 0.34$ on LoveDA with every fold positive and every class improving — **so our method composes with the nearest published competitor rather than beating it**,

taking its reproduction from 36.99 to 39.52. On OpenEarthMap it helps neither pipeline’s headline, for the same reason in both: the fit must sacrifice a catch-all sitting at 17.13 IoU that owns a ninth of the metric.

The scale reaches what thresholds cannot. It adds +4.86 on Potsdam, +1.16 on LoveDA and +5.89 on UAVid over thresholds alone. Potsdam’s **tree** is the clean case: the largest precision-recall gap we measure (+54.7), which per-class thresholds move +0.32 and the scale moves +**21.17**. The reason is measurable — **tree**’s residual is *below* the threshold but with a *different class* winning the argmax, so lowering its own threshold returns those pixels mislabelled.

Report both metrics. mIoU is an *unweighted* mean over N classes, so **a catch-all owns exactly $1/N$ of the score however meaningful it is** — 11.1% on OpenEarthMap for a class covering 0.84% of the pixels. We observe that class moving the headline in both directions: supplying +22.67 IoU of an apparent gain in one experiment (Fig. 3) and cancelling a real one in another. The distortion exists before any method is applied — OpenEarthMap’s published headline is depressed 3.38 points by that single class — so we report mIoU over the real classes beside the headline, and suggest the field does.

The correction reaches the residual the paper opened with. Held out, it lowers the discard rate 29.25% $\rightarrow$ 25.01%, and **79.2% of the pixels it recovers land on the correct class — 0.26 wrong per right, against 1.73 for the global threshold sweep that reaches the same mass**.

UAVid, and a correction to our own headline. On drone video both levers hold under flight-grouped folds and transfer across the split boundary: fitted on the training split and applied unchanged, they take 56.86 to 57.92 and then to **63.55**, verified in the unmodified pipeline against a cached prediction of 63.55 — exact on the headline, every class within 0.20. Pixel accuracy rises 79.24 $\rightarrow$ 84.81 and mean precision and recall *both* rise (+0.60, +4.92). **We then found that most of the scale’s gain there was repairing a word**, and restate the contribution accordingly in the next section.

7 A larger lever than either, and no rule for setting it

Every training-free open-vocabulary method inherits a hand-written list of class prompts, and none of the work we compare against reports how that list was chosen. Our own results above are stated at a vocabulary we did not write. We therefore tested it on two datasets, with the prediction committed to version control before each run.

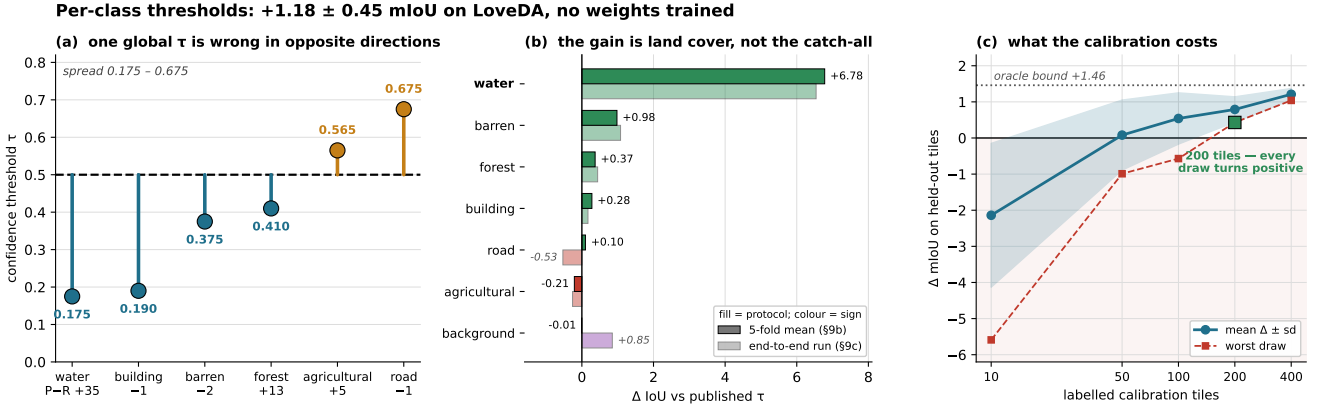


Figure 2: Left: fitted thresholds against the single published value. Centre: per-class IoU change. Right: the calibration curve — the gain is reliably positive from roughly 200 tiles on LoveDA, and from 3 flights on UAVid.

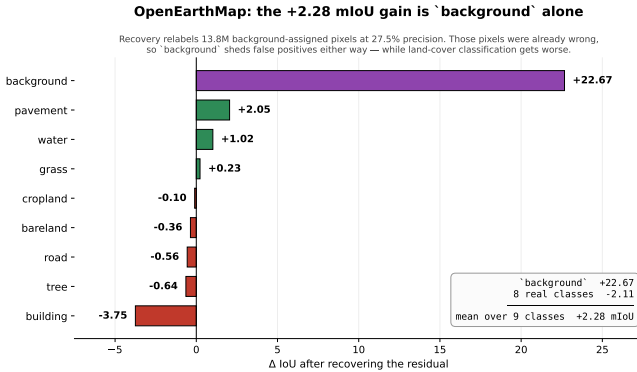


Figure 3: A headline mIoU gain that is not a segmentation improvement: the catch-all supplies $+22.67$ IoU while the real classes lose -2.11 . We observe the same artefact with the opposite sign elsewhere, which is why we treat it as a property of the metric.

On UAVid one word is worth $+3.53$ mIoU, and it is surgical. tree gains $+14.66$ and the low-vegetation class $+11.14$, while every other real class moves by at most 0.04 . The mechanism appears in the two quantities we registered in advance: the shipped vegetation was firing on anything green, and its precision rises $53.62 \rightarrow 71.51$; tree was losing those pixels, and its recall rises $55.90 \rightarrow 72.76$.

On Potsdam the identical rule costs -2.71 mIoU. We registered that the change would move the score by less than a point and were wrong about the magnitude. The shipped grass is *better* than the official term: specific, visual, and crucially it does not name the neighbouring class. So the annotation guide’s own wording is not automatically the better prompt.

Table 3: The same rule — use the dataset’s own class name — applied to the same class, on two datasets. One word changes; prompt count per class does not.

	shipped	annotation guide	Δ mIoU
UAVid	vegetation	low vegetation	$+3.53$
Potsdam	grass	low vegetation	-2.71

The prediction that did hold identifies the mechanism. Potsdam’s tree recall is 38.63 before the change and 38.63 after — unchanged to a hundredth — while its precision falls $93.34 \rightarrow 92.17$ as the weaker prompt leaks pixels into it. **Renaming a competing class changes nothing about tree there.** Its deficit is visual confusion at 5 cm ground sampling distance, not a naming error, which is why the scale reaches it and a better word does not. **The two datasets present identically in the results table — a scale gain carried by tree — and have different causes.**

The word and the multiplier are substitutes. On UAVid the two routes end in the same place: keeping the flawed prompt and letting both levers repair it reaches 63.55 , while correcting the word first and then applying both levers reaches 63.62 — 0.07 mIoU apart, and within about a point per class. **Typing two words and fitting a multiplicative reweighting on 200 labelled frames address the same pixels.** One class does not substitute: human gains the same either way, because it is not losing an argmax to a mis-named neighbour, it is simply too conservative.

We therefore restate our own result. Correcting the word removes 64% of the scale’s gain on UAVid ($+5.64 \rightarrow +2.03$), so that dataset’s method contribution is $+3.22$ — $+1.19$ from thresholds and $+2.03$ from the scale on a corrected baseline of 60.40 , reaching 63.62 .

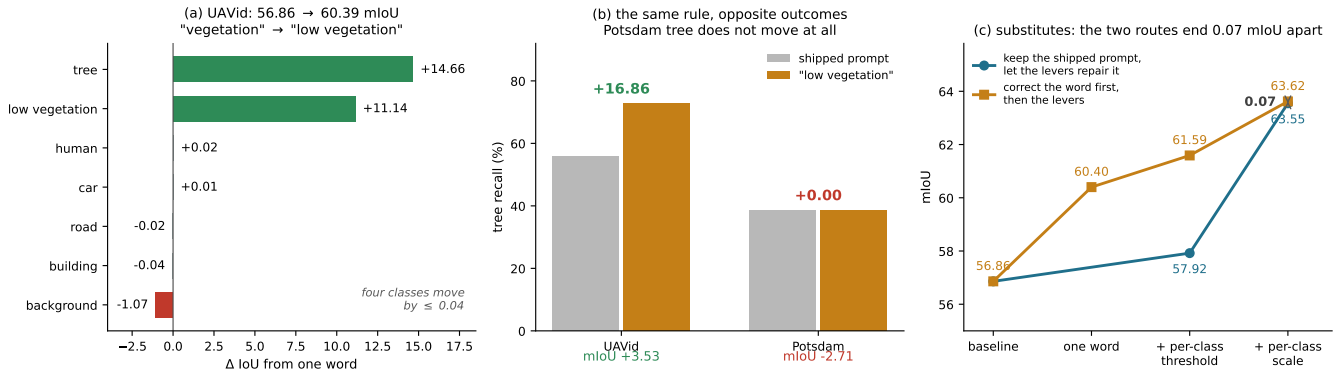


Figure 4: One word against the whole method. **(a)** On UAVid the correction is surgical: two classes move, four move by at most 0.04. **(b)** The same rule on Potsdam *costs* mIoU, and *tree*’s recall does not move at all — the two datasets present identically in a results table and have different causes. **(c)** The word and the multiplier are substitutes: correcting the prompt first, or leaving it and letting both levers repair it, end 0.07 mIoU apart.

That is in line with +2.32 on LoveDA and +5.67 on Potsdam.

What we take from this. A one-word change is worth +3.53 on UAVid and +4.94 on LoveDA, against +2.32 for our complete method on LoveDA. **The vocabulary is the largest single lever we measure, it is unreported, and the direction of a change to it is not predictable from the class name.** We do not offer a method for choosing prompts — our two arms disagree, which is the point. We offer the measurement, and the recommendation that prompt lists be reported and ablated alongside the architecture, because on these benchmarks they are worth more than the methods being compared.

7.1 Three further levers, and what their failure delimits

Per-class thresholds are everything available after the argmax, and the scale acts *at* it. That leaves the score the argmax reads, and we fitted three further per-class corrections on the same tiles, the same protocol and the same acceptance gate — mean less twice the fold standard deviation above zero, *and* every fold positive. **All three are null.**

Fusing SAM 3’s two heads per class rather than by a hardcoded maximum gives +0.04 on Potsdam and +0.22 on LoveDA. Weighting the presence score per class gives +0.16. Allowing an additive term as well as a multiplicative one — vector scaling in the sense of Kull *et al.* [4], the family a reader of the calibration literature would ask for next — gives -0.07 ± 0.18 . **We pre-registered each as a small positive and were wrong three times;** the supplementary material gives the fits and the diagnosis of each.

The pattern is worth more than any of the three. The two corrections that work reshape the *decision* —

which label a score earns, and which class wins. The three that fail reshape the *evidence* the decision reads. Each null lever nevertheless recovers the structure it was designed for: the head fit rediscovers the things/stuff duality with no supervision about which class is which, and the presence weight orders the classes by how much the gate touches them at Spearman +0.886 with exact $p = 0.017$. **They identify the right thing to correct and find that the two working levers have already collected that mass by another route.** We report them because a method claiming two corrections is stronger when the neighbouring three are measured rather than left unasked.

8 Why the calibration set is irreducible

The method costs 200 labelled tiles. It is natural to ask whether a label-free rule could choose the thresholds instead, and we tried eleven — Otsu splits, confidence percentiles, presence-scaled thresholds, cross-head agreement, and others. None reaches the oracle bound; on LoveDA every one scores *below* the published threshold.

The instructive part is why. We had supposed the missing quantity was per-class precision, which is label-derived by definition. It is not: four of the label-free statistics predict per-class precision at Spearman $\rho \geq +0.886$ with exact $p \leq 0.033$, and they still buy nothing. Precision does not determine the threshold, and the relation is not even monotone — *water* at 88.5% precision takes the *lowest* threshold while *road* at 68.5% takes the *highest*.

The real bound is structural. The optimal threshold vector solves a *coupled* multi-class objective: raising one class’s threshold pushes pixels into the catch-all, changing every other class’s optimum. No per-class scalar can express that, measurable or not. **That is why a six-parameter fit works and every one-parameter rule**

fails, and it generalises to any training-free pipeline that thresholds a per-class score.

9 Limitations

Our remaining results are stated at vocabularies we have not audited. Sec. 7 shows one word is worth more than our complete method on LoveDA and that the direction of such a change is unpredictable. We tested two prompts on two datasets. LoveDA’s and OpenEarthMap’s lists are untested by the same rule and we cannot exclude that part of their reported gains is available by rewording. Two things bound this rather than excuse it: Potsdam was tested *because* its scale gain has UAVid’s signature and came back clean, and LoveDA’s list already carries hand-tuned synonyms whose removal costs +4.94 mIoU. Neither substitutes for auditing it.

One reproduction does not reproduce. Our UAVid baseline is +2.16 above the published figure under identical code, configuration, vocabulary and thresholds, and their released configuration points at a split whose labels are not public. We quote gains against our own reproduction throughout, as we do in the opposite direction for ConInfer, whose LoveDA figure we reproduce 2.34 *below* its published value. A difference measured on one pipeline is robust to a constant offset in it; an absolute comparison across two is not, and we do not make one.

Thresholds do not transfer across a domain shift. Fitted on LoveDA’s rural tiles and applied to urban ones they land *below* the published threshold. The calibration set must match the evaluation distribution, and pooling across two distributions reproduces the failure of the global threshold one level up. We find one label-free predictor of when transfer will work — agreement in discard rate between source and target, which orders four split pairs — but four points across two datasets is an ordering, not a law.

Seventy frames from seven flights is a sample of seven. UAVid’s validation split is video, so its effective sample size is the flight count; a five-fold inside it has a baseline that swings eighteen mIoU points across folds. We fit on the training split and evaluate on validation whole for that reason. The same caution applies to any video-derived benchmark evaluated with frame-level splits.

The negative results are ours too. Three further per-class corrections — fusing the two SAM 3 heads, weighting the presence score, and affine rather than diagonal scaling — were fitted under the same protocol and gate, and all three are null (the supplementary material). Each was pre-registered as a small positive.

10 Conclusion

A single confidence threshold is the wrong shape for a multi-class open-vocabulary decision. Fitting one per class, and one multiplier per class before the argmax, recovers a consistent gain across four datasets and two backbones on the supervision the baseline already spends — and the first of those two rules is provably everything available after the argmax. But the largest lever we found is not ours: the hand-written prompt list that every method in this literature inherits and none reports is worth more, on these benchmarks, than the methods being compared. We recommend it be reported.

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